

EXPERIENCE

- PhD Researcher

Full-time · 30 h/wk

RWTH Aachen University · Germany

Aug 2023 – Present

"Sparse Exit Wave Reconstruction via Deep Unfolding"

Incorporating data-driven learning into model-based exit wave reconstruction, a variant of phase retrieval, using deep unfolding to create new hybrid approaches.

Phase Retrieval · Image Processing · Optimization · Deep Neural Networks
- E-Learning & Assessment Assistant

Mini-job · 10 h/wk

RWTH Aachen University · Germany

Apr 2026 – Present

Managing exercises and online assessments via Dynexite, publishing materials on Moodle, and handling student grading and feedback.

Dynexite · Moodle · OnLine Assessment · Grading
- Research Internship & Master's Thesis

Part-time · 20 h/wk

Zuse Institute Berlin · for the M.Sc. at TU Berlin

Sep 2022 – May 2023

"Approximation of Koopman Operators Using Random Basis Functions"

Implemented Extended Dynamic Mode Decomposition with random Gaussians and neural networks in Python to investigate Koopman operator theory and nonlinear dynamical systems.

Koopman Operators · Neural Networks · Dynamical Systems
- Research Internship & Master's Thesis

Full-time · 35 h/wk

Université de Bordeaux · for the M.Sc. at UPPA, France

Feb – Aug 2020

"Semi-discrete Optimal Transport for Large-Scale Problems"

Implemented gradient descent algorithms in Python to solve large-scale semi-discrete optimal transport problems in image processing and machine learning.

Convex Duality · Image Modeling · Gradient Descent
- Master's Thesis — Control Theory

Lebanese International University · Lebanon

Feb – Aug 2019

"Decay Rates of a Timoshenko System with Kelvin-Voigt Damping"

Proved that the energy of the Timoshenko system with Kelvin-Voigt damping dissipates exponentially or polynomially depending on the material coefficient.

C0-Semigroup · Timoshenko Beam · Polynomial & Exponential Stability

SEMINARS & PROJECTS · FOR THE M.Sc. AT TU BERLIN

- Seminar — Optimal Transport

"Entropic Regularization of Discrete and Continuous Optimal Transport"

Studied Sinkhorn's algorithm, the dual formulation, and convergence under small and large regularization.

Sinkhorn's Algorithm · Dual Formulation · KL-Divergence
- Seminar — Nonlinear Optimization

"Constrained Optimization and Duality — Applications to ML"

Lagrangian duality and constraint qualifications, formulating the SVM dual, and coordinate and gradient descent methods.

Lagrangian Duality · SVM Dual · Coordinate & Gradient Descent
- Project — Scientific Computing

"Gas Transport in Porous Medium"

Numerical solution of the nonlinear Porous Medium Equation using the Voronoi Finite Volume Method in Julia, with 1D and 2D simulations.

Porous Medium Equation · Voronoi FVM · Julia

PUBLICATION

- "A generalization bound for exit wave reconstruction via deep unfolding"

Preprint · related to PhD – [arXiv 2025](#)

EDUCATION

- M.Sc. Scientific Computing

Technische Universität Berlin

2021 – 2023

GPA 1.2 · Excellent
- M.Sc. Mathematics, Modeling & Simulation

Université de Pau et des Pays de l'Adour

2019 – 2020
- M.Sc. Applied Mathematics

Lebanese International University

2018 – 2019

Rank 1 · GPA 4.0 · Excellent
- Master 1, Pure Mathematics (1st Year of Master)

Lebanese University

2014 – 2015
- B.Sc. Pure Mathematics

Lebanese University

2008 – 2012

ACHIEVEMENTS

- Received **A** in the research internship, master's thesis, and most courses of the M.Sc. at TU Berlin.

Received **A** in all required courses of the M.Sc. at LIU.

Excellence Scholarship from Région Nouvelle-Aquitaine to pursue the master's at the University of Pau.

Ranked 1st in the Number Theory course, Master 1 at LU.
- SKILLS
- Scientific & Mathematical
- Optimization

Machine & Deep Learning

Deep Unfolding

Numerical Simulation

Optimal Transport

Neural Networks

Bayesian Inference

PDEs · FEM / FVM
- Computing
- Python

TensorFlow

Matlab

Git

PyTorch

Julia

OpenMP/MPI

LaTeX
- Personal
- Teamwork & Collaboration

Adaptability

Problem Solving

Fast Learner
- LICENSES & CERTIFICATES
- Machine Learning Specialization

Coursera · DeepLearning.AI & Stanford

2024

Supervised Learning · Advanced Algorithms · Unsupervised Learning & RL
- MATHEMATICS TEACHER
- Private Schools · Beirut, Lebanon

2012 – 2018

Taught mathematics in English at intermediate and secondary levels, with private tutoring.
- LANGUAGES
- Arabic Native English C1 German B1 French A1